

AI-powered virtual streamers and viewer behavior: An image-inspiration-behavior framework

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ABSTRACT

Today, a new form of live-streaming has emerged: virtual streamers powered by Artificial Intelligence technology, which autonomously perform live-streaming tasks as human streamers do. This form of live-streaming has quickly gained popularity. However, studies in this area are scarce. This study aims to fill this research gap by investigating how virtual AI streamers impact viewers by adopting the image-inspiration-behavior framework, in which the image of virtual AI streamers is described by warmth, competence, and coolness, and behavior is studied as interaction and purchase intentions. The data were collected from 559 participants via a scenario-based survey. The partial least squares – structural equation modeling and artificial neural network are used to analyze the data. It was found that the three image dimensions (warmth, competence, and coolness) influence consumer interaction and purchase intentions through the mediating effect of inspiration. The findings of this study advance previous studies mainly by investigating how virtual AI streamers impact viewer behaviors and offer insights to live-streaming companies in their adoption of virtual AI streamers.

1. Introduction

The popularity of live-streaming e-commerce (LSE) has reshaped traditional e-commerce and has become an important retail promotion channel [1,2]. Compared with traditional e-commerce promotions, the uniqueness of LSE lies in the “watching while purchasing” function and real-time interaction between live streamers and viewers [3]. Today, a rather new phenomenon in LSE has emerged: live streamers powered by artificial intelligence technology (hereinafter referred to as “virtual AI streamers”) [4].

Unlike simple virtual streamers which are digitally designed characters controlled by real humans who often remain anonymous [5], virtual AI streamers are digitally created artificial humans who can autonomously perform human tasks using algorithms [6–8]. In other words, virtual AI streamers interact with their viewers in a human-like manner and express emotions like humans do without backstage human control. For example, Neuro-sama, an English-speaking virtual AI streamer, appears in the form of an anime girl and plays Minecraft and Osu (a musical rhythm game) during live-streaming; while playing games, Neuro-sama interacted with over 50,000 global viewers by responding verbally to comments made in the chat, singing songs according to the requirements of audiences, and even displayed a “learning” process by improving its performance such as gaming and singing skills based on audience feedback and making interactions more

responsive to the audience’s demonstrated preferences [9].

This form of LSE has been adopted by many retailers today for its popularity among viewers and advantages of virtual AI streamers, e.g., working 24*7 without fatigue, accuracy in answering questions, and stable emotion [8,10]. The recent advancements in digital human large language models (LLMs) have further accelerated this trend by significantly reducing the creation cost of virtual AI humans, now achievable for as little as \$145 [11]. During a recent seasonal promotional campaign by JD.com, China’s largest retailer, approximately 4,000 brands created their own virtual AI streamers, highlighting the rapid adoption of this technology [12]. As a result, virtual AI streamers have shown great commercial prospect: according to the Annual Research and Consultation Report of Panorama Survey and Investment Strategy on China Industry [13], virtual AI streamers can achieve conversion rates 2.3 times higher than traditional streamers and are projected to generate more than 30 percent of Gross Merchandise Value by 2026 in China.

Despite the popularity of virtual AI streamers, theoretical research lags far behind practical applications. To our knowledge, only Feng et al. [14], Wang et al. [8], and Gao et al. [10] have studied how virtual AI streamers influence viewer responses. Feng et al. [14] found virtual AI streamers satisfy viewers and motivate them to purchase by arousing positive emotions. Wang et al. [8] compared live-streaming effectiveness between human and virtual AI streamers and found they generate

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different sales volumes and engagement in different product categories. Gao et al. [10] revealed the effect of likeability, animacy, and responsiveness on purchase intention via the mediation of presence. The three studies make noticeable contribution to this area; however, they may not be sufficient. To advance knowledge in this area, we examine important viewer behavioral responses in LSE (i.e., interaction and purchase intentions) [15] and their influencing factors.

Anthropomorphism is one of the most widely studied factors behind consumer response towards AI technology, since human likeness is an important and unique technological characteristic of AI [16–18]. In this study, we do not intend to examine anthropomorphism by studying the degree of human likeness as most previous studies do [19]. Instead, we examine anthropomorphism from a different perspective by studying a combination of image dimensions that describe the human-likeness characteristic, i.e., warmth, competence, and coolness. Warmth and competence are generally used to measure the image of human social media influencers [20,21], and coolness, commonly applied to assess people's creativity, autonomy, and attractiveness [22], maps onto the technological novelty that sets virtual AI streamers apart from human counterparts [23]. The application of established human image dimensions on virtual AI streamers signals their anthropomorphism [24]. Thus, these three image dimensions are adopted here to capture the human-likeness characteristic of virtual AI streamers. In addition, coolness is a higher-order social impression influenced by trait-level image dimensions such as warmth and competence [25]. Therefore, this study investigates the respective contributions of warmth and competence to coolness so as to clarify the internal structure among the anthropomorphized image dimensions.

Virtual AI streamers are new and creative concepts to most people [10,14], and thus their potentially warm, competent, and cool image is capable of stimulating viewers' interest and imagination and inspiring them to act accordingly [26], such as interacting with virtual AI streamers and purchasing the products they promote [27]. Therefore, inspiration is incorporated in this study as a mediator of the relationship between the image of virtual AI streamers (i.e., warmth, competence, and coolness) and viewer behavioural responses (i.e., interaction and purchase intentions). Inspiration has been examined in previous studies of new technologies such as augmented reality (AR) [28,29], since new technologies facilitate the creation of inspiring content [30]. Therefore, inspiration can help explain viewer responses in the context of the new technological context of virtual AI streamers.

Consequently, the objective of this study is to investigate how virtual AI streamers impact viewers by adopting the image-inspiration-behaviour framework, in which the image of virtual AI streamers is described by warmth, competence, and coolness, and behaviour is studied as viewer interaction and purchase intentions. In particular, the specific objective includes: 1) to examine the potential relationships among the three image dimensions of virtual AI streamers; 2) to investigate the influence of the three image dimensions on inspiration; 3) to study the effect of inspiration on interaction and purchase intentions; 4) to understand the mediating effect of inspiration. The findings of this study will contribute to the theoretical development in the following aspects: 1) adding to the emerging literature on virtual AI streamers by understanding how such streamers impact viewers; 2) studying anthropomorphism from a different perspective, i.e., adopting three image dimensions to depict the human-likeness characteristic instead of focusing on its degree; 3) advancing the knowledge of inspiration by incorporating virtual AI streamers into the inspiration discussion. The findings of this study will also offer guidance for practitioners including LSE companies about how to facilitate desired viewers' behaviour by adjusting certain characteristics of virtual AI streamers (e.g., by putting more emphasis on which image) and also retailers interested in using virtual AI streamers in marketing and promotion.

2. Literature review

2.1. Conceptual framework of image-inspiration-behavior

To understand how virtual AI streamers impact viewers, this study adopts the framework of image-inspiration-behavior. The key behavioral responses of interaction and purchase intentions are examined given their importance in LSE as already mentioned [31].

The framework is primarily grounded in dual-system theories. Dual-system theories explain behavioral responses as a result of two distinct systems: System 1 is fast, intuitive, and emotional, while System 2 is slow, deliberate, and analytical [32]. Given the fast-paced and hedonic nature of LSE [33], especially in the case of virtual AI streamers whose content focuses on talent performance and entertainment [34], viewers in the context of this study are more likely to rely on intuitions as mental shortcuts to simplify decision-making (System 1). Under such conditions, viewer behavior is primarily influenced by streamer-related factors, such as streamer image [35]. Therefore, the anthropomorphized image of virtual AI streamers is adopted as stimulus here, which is represented by the traditional human streamer image of competence, warmth, and coolness.

Complementing this cognitive approach, motivational salience theory further explains why the anthropomorphized image (competence, warmth, and coolness) functions as a psychologically salient stimulus in this study [36]. The competent, warm, and cool image of human streamers are typically stimuli associated with rewards to viewers, e.g., pleasure and customer satisfaction [37]. When viewers encounter novel yet anthropomorphized virtual AI streamers, they anticipate a rewarding outcome from previously learned reward cues and thus involve a motivational urge for immediate consumption [38], which is aligned with intuitive decision-making (System 1).

This study adopts inspiration as a motivational mediator between image stimulus and behavioral responses. Inspiration is a motivational state spontaneously evoked by sources characterized by new ideas, imaginative appeal, and approach elicitation rather than avoidance [39]. Inspiration is well-suited to the context of virtual AI streamers, whose technologically novel yet human-like image stimulates viewers' imagination and whose design often aims to encourage interaction and purchase behavior from viewers [40]. Prior studies have established the mediating role of inspiration between appealing social media content and customer behavioral intentions [41]. Therefore, inspiration, which both captures the intuitive nature of viewer processing (System 1) and aligns with the characteristics of this context, is adopted as a motivational bridge between image stimulus and behavioral responses.

2.2. Image of virtual AI streamers

Competence and warmth "form basic dimensions that, together, account almost entirely for how people characterize others" [42]. Competence is the evaluation of people's abilities such as intellect, skill, inventiveness, and effectiveness, and individuals of competence could be described as brilliant, creative, efficient, ahead of the curve, inventive, intellectual, and informed [20]. Warmth relates to characteristics of friendliness, helpfulness, sincerity, honesty, and morality [42]. While competence and warmth are used to evaluate humans, they could also be used to judge AI agents [43,44]. In the domain of AI studies, competence is defined as the extent of underlying knowledge that an AI agent demonstrates [45]. When an AI system is asked to display its knowledge or behave under certain conditions, its competence appears as its performance [45]. When an AI agent shows good intent toward others by displaying appropriate emotions (e.g., glad to help people in need), it was more likely to be recognized with warmth [46], and robots' human-like appearance and postures are also associated with warmth [47].

Coolness is a term commonly used to assess the attractiveness of things or experiences, which also shows exceptional, superior, and exclusive qualities [48,49]. More importantly, appropriate autonomy,

which refers to diverging from beliefs, norms, and others' expectations, underlies coolness and differentiate it from other similar concepts such as product quality [50]. The coolness of new technology is explained as innovativeness and creating new things rather than sticking with the old ones [23].

Coolness theory suggests that perceived coolness is developed through two routes: dominance (reflecting independence and outside of social constraints) and social desirability (reflecting qualities that are valued, desired, and liked) [25]. While dominance is the quality often admired in cultural icons and trendsetters, social desirability ensures the quality is positively received [50]. The two routes closely align with competence and warmth from the Stereotype Content Model [42]. Specifically, competent people or things are perceived as capable and independent, thus signalling autonomy and dominance which contribute to coolness; warm people or things, being friendly and kind, are more likely to be associated with social desirability pathway to coolness [51]. In the context of virtual AI streamers, competence is manifested in their ability to provide immediate and knowledgeable feedback, highlighting their independent capabilities and intellectual autonomy. Warmth is reflected through their human-like friendliness and interaction, enhancing their social appeal and acceptance among viewers. Given these theoretical considerations, the following hypotheses are proposed:

H1. *Competence is positively related to coolness.*

H2. *Warmth is positively related to coolness.*

2.3. Inspiration

Inspiration can be understood as a "motivational state that compels individuals to bring ideas into fruition" [52]. The key characteristics of inspiration are evocation, transcendence, and approach motivation [53]. Evocation is the fact that inspiration is spontaneously evoked by external stimuli. Transcendence describes individuals' state of obtaining new awareness or thoughts, like self-enhancement. The inspired individuals are then motivated to actualize their new ideas, which is approach motivation.

In the domain of consumer studies, inspiration is studied as customer inspiration, which refers to a "customer's temporary motivational state that facilitates the transition from the reception of a marketing-induced idea to the intrinsic pursuit of a consumption-related goal" [39]. Sources of marketing-induced ideas include advertisements, in-store presentations, visual features of websites [54], brand experience [26], travel-related posts in social media [30], and many more. Today, marketing-induced ideas have been expanded to new technologies such as AR and include sources such as augmentation quality [28,29]. Virtual AI streamers' human-like impressions of competence, warmth, and coolness via autonomously performing human streamers' tasks are new and creative to viewers and appeal to viewers to imagine new possibilities beyond limits [8], which are in line with the evocation and transcendence characteristics of inspiration [53,55] and serve as a necessary condition for inspiration to occur [39]. Consequently, the following hypotheses are proposed:

H3. *Competence is positively related to viewer inspiration.*

H4. *Warmth is positively related to viewer inspiration.*

H5. *Coolness is positively related to viewer inspiration.*

Customer inspiration is a powerful construct capable of influencing customer behavior, including customer engagement [56], loyalty [57], and purchase intention [58]. Customer inspiration acts like intrinsic motivation to actualize new possibilities and offers a mental shortcut to simplify the rational decision process [30], and the resulting behaviors depend on the inspirational content [39,59], which is in line with the key characteristic of approach motivation. Customer inspiration could affect behaviors from two aspects: social glue and choice imitation [58].

Social glue is the emotional tie between the suppliers of inspiration (virtual AI streamers) and receivers (viewers), which facilitates viewers to interact with the virtual AI streamers and other viewers; choice imitation means receivers are inspired to make the same choice as suppliers, such as purchasing the same products that the virtual AI streamers promote. Consequently, the following hypotheses are proposed:

H6. *Viewer inspiration is positively related to interaction intention.*

H7. *Viewer inspiration is positively related to purchase intention.*

2.4. Interaction and purchase intentions

Customer interaction intention refers to the extent to which customers are willing to exercise reciprocal influence over other communication parties during service encounters [60,61], which is often studied as customers' willingness to exchange information and feelings with other parties [62]. Customer interaction has benefits including acquiring information, strengthening ties with others, enhancing their confidence and perceived status, and being rewarded and relaxed [63]. Therefore, viewers' real-time interaction during live-streaming allows them to gain a better-tailored comprehension of the product [15], and to experience more excitement and enjoyment from social interaction in public online spaces and less uncertainty and hesitation from getting pre-sales information [64,65]. Thus, interaction during live-streaming boosts viewer purchase intention [66,67]. Consequently, the following hypothesis is proposed:

H8. *Interaction intention is positively related to purchase intention.*

2.5. Mediating effect of viewer inspiration

Based on the above, the model of all hypotheses is shown in the Fig. 1. The image dimensions of virtual AI streamers (warmth, competence and coolness) facilitate viewer inspiration; viewer inspiration fosters interaction and purchase intentions, and since the core in this study is viewer inspiration, the following hypotheses were proposed:

H9. *The effect of competence (a), coolness (b), and warmth (c) on interaction intention is positively mediated by viewer inspiration.*

H10. *The effect of competence (a), coolness (b), and warmth (c) on purchase intention is positively mediated by viewer inspiration.*

3. Methodology

3.1. Scenario design

A scenario-based survey was used in this study, which is suitable for research on virtual AI streams [7,68]. Hua Zhibing, the first Chinese virtual AI streamer, was chosen for the scenario in this study. Designed with high IQ and EQ, she can create scripts, draw, compose poetry, and think logically and interact emotionally, who differs from other virtual AI agents as she is driven by the cutting-edge Enlightenment 2.0 learning model [34].

The participants in the scenario were instructed to read a brief introduction about Hua Zhibing, including both character images and capability descriptions, and then watch a video of her singing while playing guitar. Subsequently, the participants were informed that Hua Zhibing would promote and sell products through live-streaming. Participants were told to imagine themselves watching the live-streaming of her, learning about products and business activities from her, and participating in the interactions during the live-streaming. The design focused on talent performance instead of product sales, mirroring real-world virtual AI streamers such as Neuro-sama, whose content centers on entertainment, interaction and talent display, with product mentions kept brief, as sales depend more on the emotional connection with

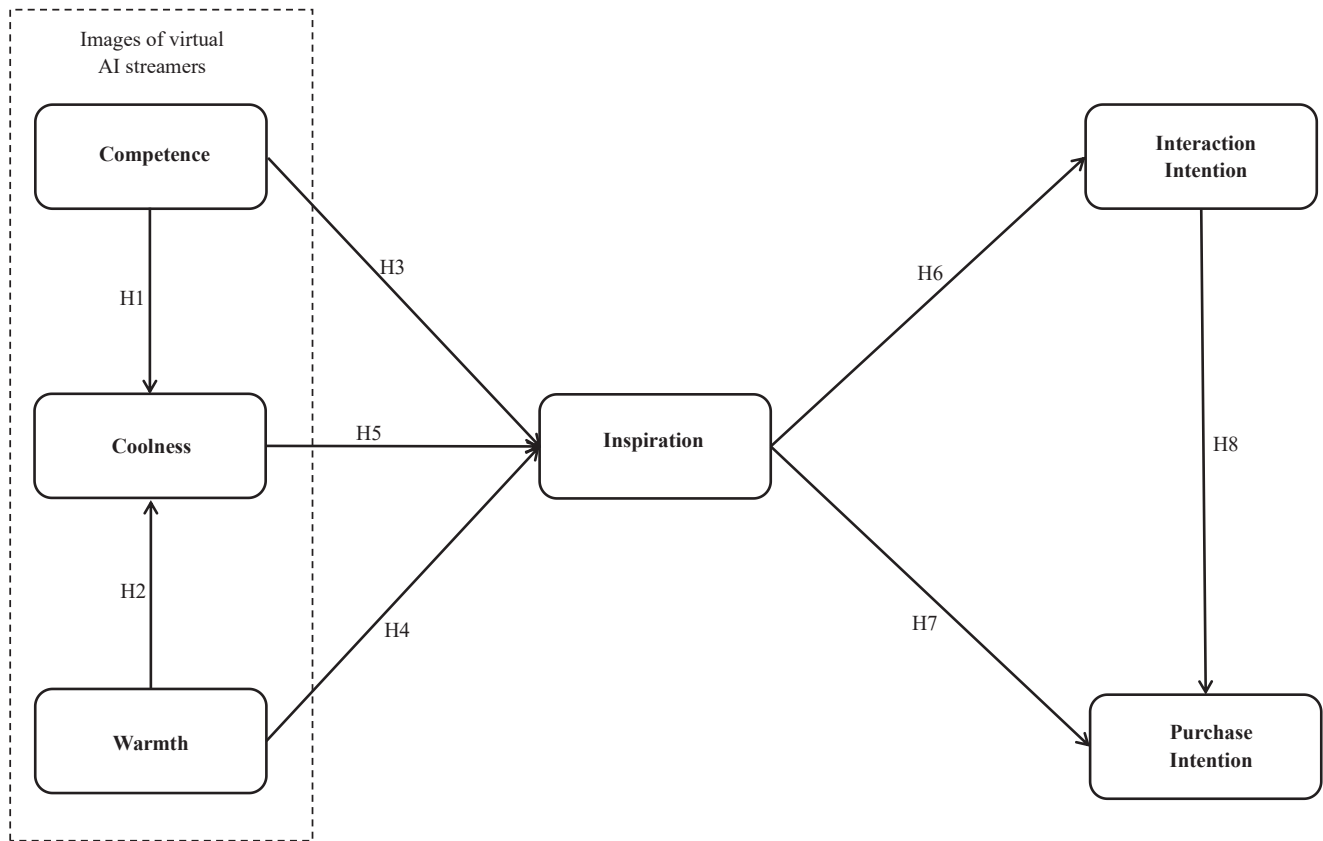


Fig. 1. Proposed research model.

viewers rather than explicit product promotion [69]. Her posters and pictures of the live-streaming were also displayed to the participants, helping them better imagine the scenario. To avoid any potential external noise, neither specific product nor brand information was provided. Finally, they were asked to answer the questionnaire surveys. Appendix 1 shows the images presented to the participants.

3.2. Measurement

The questionnaire's constructs were taken from previous research and modified for the LSE context in this study. The items for perceived competence and perceived warmth were from Fiske et al. [20]. A four-item scale for perceived coolness was developed from Sundar et al. [70] and Liu et al. [21]'s sub-factor of coolness and originality, which measures the extent of uniqueness and novelty. The five items of consumer inspiration in Böttger et al. [39] were used to measure inspiration. Four items were modified from Kim et al. [71] and Shi et al. [72] to measure interaction intention. Using four items from Beatty and Ferrell [73], purchase intention was measured. Each item was rated with a seven-point Likert-type scale, with 1 point denoting "strongly disagree" and 7 points denoting "strongly agree". The survey also included questions on background information at the end, such as gender, age, education, monthly income, and prior understanding of virtual AI streamers. One of the writers, who is also a certified Chinese-English translator, translated all of the English items using the back-translation approach [74]. In order to deal with ambiguous issues and review the scenario settings, five university students who were fluent in both Chinese and English were employed to discuss their doubts and opinions about each question. Ultimately, the measurement items' readability, articulateness, and accuracy were enhanced.

3.3. Data collection

A method based on quantitative research was employed to test the hypothesized model in this study, and the data were collected via a self-administered questionnaire. A pilot study was carried out prior to the questionnaires being officially distributed, during which 74 valid responses were used to increase the reliability and validity of the questionnaire. After this phase, the questionnaire was formally distributed with a simple random sampling approach using the specialized data collection services of <https://www.wjx.cn>, one of the largest online survey platforms in China. Participants were provided with monetary rewards. This study's target respondents are viewers who have previously experienced live-streaming shopping, since their previous purchase experience could help them imagine a virtual AI live-streaming scenario. Therefore, a screening question was used to disqualify participants without experience with live-streaming purchases, and they were only shown the scenario if they passed the filtering question. Furthermore, the survey included two manipulation check questions: "I never participated in livestream commerce" and "Her name is Huahua." The responses that agreed with these two questions were deleted.

Between December 28, 2021, and January 13, 2022, empirical data was collected. During the two-week period, 1,782 participants participated in the survey. 559 samples were considered valid, with a valid response rate of 31.4 %. The sample-to-variable ratio suggests a minimum participant-to-variable ratio of 5:1 and a preferred ratio of 15:1 or 20:1 [75]. Thus, the sample size adopted for this study is supported by theoretical and statistical evidence. Table 1 shows the participants' demographic information. The demographic distribution is consistent with the patterns in the prior live-streaming literature (e.g., Li et al. [76]; Lu and Chen [77]).

Table 1

Demographic information (n = 559).

Measure	Item	Frequency	Percentage (%)
Gender	Male	231	41.3
	Female	328	58.7
Age	18–24	56	10.0
	25–30	289	51.7
	31–35	148	26.5
	36–40	49	8.8
	41–50	16	2.9
	50 and above	1	0.2
Monthly income	<3,000 RMB	19	3.4
	3,000–5,000 RMB	63	11.3
	5,001–10,000 RMB	304	54.4
	>10,000 RMB	173	31.0
Education	High school or below	4	0.7
	College	471	84.3
	University	51	9.1
	Postgraduate or higher	33	5.9
Prior knowledge of virtual AI	Yes	487	87.1
	No	72	12.9

3.4. Data analysis

The data analysis was conducted using the SPSS 27 and WarpPLS 8.0 program. Initially, the SPSS 27 program was used to display background information about the participants. Then, we conducted the exploratory factor analysis (EFA) with SPSS 27. Following that, the reliability and validity of the questionnaire were evaluated, after which the hypothesized model was tested using the PLS-SEM approach by the WarpPLS 8.0 program. As a “causal-predictive” approach to SEM, PLS has been used to avoid the apparent conflict between explanation and prediction [78]. PLS-SEM was selected because of its predictive capabilities and accurate performance in non-normal data [79].

Additionally, in order to assess each independent variable’s predictive power, an artificial neural network (ANN) was generated using Python in this study. The utilization of ANN has been widespread in anticipating consumer psychology and behavior for an extensive period (e.g., Badea (Stroie) [80]; Gan et al. [81]; Sobhanifard and Sadatfarizani [82]; Zhou et al. [83]). By recognizing both linear and non-linear relationships between variables, this method has the potential to provide deeper insights into the anticipated linkages [84]. It is observed to forecast outcomes more robustly compared to SEM [85,86], by solving the shortcomings of other research approaches such as linear regression and PLS-SEM [87–89]. Therefore, we employed ANN in addition to SEM to assess the robustness and predictive strength of our findings from complementary perspectives.

Using an input-hidden-output layer (IHOL) architecture and a deep learning method, the Rao and Ali [90]’s technique was used to implement the ANN. Training and testing data were used for the testing process [91]. The optimizer used in this study was stochastic gradient descent (SGD), and the activation function for both hidden and output neurons was the rectified linear unit (ReLU) function. Moreover, the root-mean-square error (RMSE) was applied to calculate the loss.

4. Results

4.1. Exploratory factor analysis of virtual AI streamer image

Since there may have been an overlap in the scale elements, EFA was performed on the 12 items of the virtual AI streamer images to determine the underlying image dimensions. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was 0.82, and the correlation matrix’s overall significance was less than $\alpha = 0.001$, indicating a suitable sample size and the data’s suitability for factor analysis. The subsequent

analysis extracted three factors with eigenvalues greater than 1. However, one item had low factor loadings (<0.50) and was removed before the EFA was re-estimated [92]. In this case, the EFA produced an 11-item, three-factor model (as Table 2 displays) which could explain 58.77 % of the data variance. The first factor (F1), labeled “warmth,” is associated with viewers’ perceptions of the characteristics of virtual AI streamers, such as friendliness, helpfulness, sincerity, honesty, and morality, explaining 23.30 % of the variance. The second factor (F2) is most associated with viewers’ perceptions about the attractiveness of virtual AI streamers, such as exceptional, superior, and exclusive qualities, and is labeled “coolness,” explaining 18.82 % of the data variance. The third factor (F3) is associated with virtual AI streamers’ competence and ability viewers can perceive, such as intellect, skill, inventiveness, and effectiveness, and is labelled “competence,” explaining 16.67 % of the variance.

4.2. The measurement model

The convergent and discriminant validity of the constructs were checked using the WarpPLS 8.0 program [93]. All of the items had factor loads higher than 0.5 and were strongly connected with their constructs as they were all well-established and mature scales. Table 3 shows the means of the factors. The means of competence (5.87), warmth (5.29), coolness (5.99) and inspiration (5.51) were high, which indicated that participants had a positive perception and attitude towards the virtual AI streamer and were inspired by watching the live-streaming. Moreover, the means of interaction (5.78) and purchase intention (4.71) were also high, illustrating that participants had a high intention to interact during the live-streaming and a relatively high product purchase intention. The results had enough standard deviation, indicating sufficient variability for further investigation. In addition, the results of the Robust Jarque Bera test indicated that the components were not normally distributed. Given the lack of normality, the PLS-SEM, which was less impacted by non-normality, was suitable for analysing the data [79].

Cronbach’s alpha and composite reliability (CR) were used to measure internal consistency. Table 3 shows that all of the Cronbach’s alpha and CR coefficients were above 0.67 and 0.8, respectively, which met the threshold of Kock and Lynn [94] that at least one of these two coefficients should be not less than 0.70. As a result, all constructs were reliable. Average variance extracted (AVE) scores were used to evaluate convergent validity. All AVE scores were more than 0.50 (from 0.51 to 0.71), indicating high convergent validity [95]. Using the method

Table 2

Factor loadings for the image scale.

Items	F1: Warmth	F2: Coolness	F3: Competence
I felt that the virtual AI streamer was sincere	0.81		
I felt that the virtual AI streamer was good-natured	0.80		
I felt that the virtual AI streamer was tolerant	0.74		
I felt that the virtual AI streamer was warm	0.72		
This virtual AI streamer is original		0.77	
This virtual AI streamer is unique		0.75	
This virtual AI streamer is out of the ordinary		0.64	
This virtual AI streamer is novel		0.60	
I felt that the virtual AI streamer was competent			0.80
I felt that the virtual AI streamer was intelligent			0.74
I felt that the virtual AI streamer was confident			0.68
Eigenvalues	3.83	1.50	1.13

Table 3

The hypothesized path coefficients.

Hypothesis	Path	Coefficient	p-value	95 % CI Lower Bound	95 % CI Upper Bound	Effective Size	Results
H1	Competence → Coolness	0.30	<0.001	0.22	0.38	0.12	Supported
H2	Warmth → Coolness	0.22	<0.001	0.14	0.30	0.08	Supported
H3	Competence → Viewer inspiration	0.16	<0.001	0.08	0.24	0.06	Supported
H4	Warmth → Viewer inspiration	0.21	<0.001	0.13	0.30	0.08	Supported
H5	Coolness → Viewer inspiration	0.27	<0.001	0.19	0.35	0.11	Supported
H6	Viewer inspiration → Interaction intention	0.44	<0.001	0.36	0.52	0.22	Supported
H7	Viewer inspiration → Purchase intention	0.34	<0.001	0.26	0.42	0.17	Supported
H8	Interaction intention → Purchase intention	0.33	<0.001	0.25	0.41	0.16	Supported

proposed by Fornell and Larcker [95], discriminant validity was evaluated. As the square roots of the AVE scores for each two constructs were higher than their correlation, there was a high discriminant validity. All of the construct pairings in our investigation satisfied this requirement. Additionally, every HTMT ratio (varying from 0.38 to 0.66) was below 0.85 [96]. Ultimately, all full collinearity variance inflation factor (VIF) values (from 1.54 to 1.97) were less than 5, indicating an acceptable discriminant validity for PLS analysis.

4.3. The structural model

We used WarpPLS 8.0 to test the hypothesized model, and in order to reduce confounding effects that may result in bias, we included gender, age, income, education, and previous experience with virtual AI streams as control variables in the model. The R^2 coefficients for coolness (0.22), viewer inspiration (0.27), interaction intention (0.27), and purchase intention (0.34) are shown in Fig. 2. Thresholds of R^2 value that Chin [97] proposed are 0.67, 0.33, and 0.19 for strong, moderate, and weak, respectively. Within behavioral research, an R^2 value of 0.20 is often considered satisfactory. The R^2 values in this study showed the structural model's predictive accuracy.

All hypotheses in this study are supported, and the statistical significance was demonstrated by the one-tailed P values for path

coefficients. Significant indirect effects were also supported for each of the mediation hypotheses. Additionally, by using the stochastic instrumental variables method [93], we confirmed that the endogeneity problem did not exist in viewer inspiration and purchase intention, as the path coefficient for the relationship between the endogenous latent variable (LV) and the stochastic instrumental variable was not statistically significant with a two-tailed P value greater than 0.05 (two-tailed P values equalled 0.56 and 0.68, respectively). Fig. 2 presents the final model, and Table 4 and Table 5 provide the details of the tested relationships, including the effect size results and 95 % confidence intervals.

4.4. Artificial neural network

To investigate how independent factors affect purchase intention and interaction intention, an ANN analysis was conducted. In Fig. 3, the predictive model is displayed.

With RMSE = 0.983, the IHOL architecture with one hidden layer predicted interaction intention and purchase intention with a relatively good degree of accuracy when three independent variables—competence, coolness, and warmth—were taken into account. The mediating effects of inspiration as well as the impacts of the independent factors on the dependent variables were simulated and further

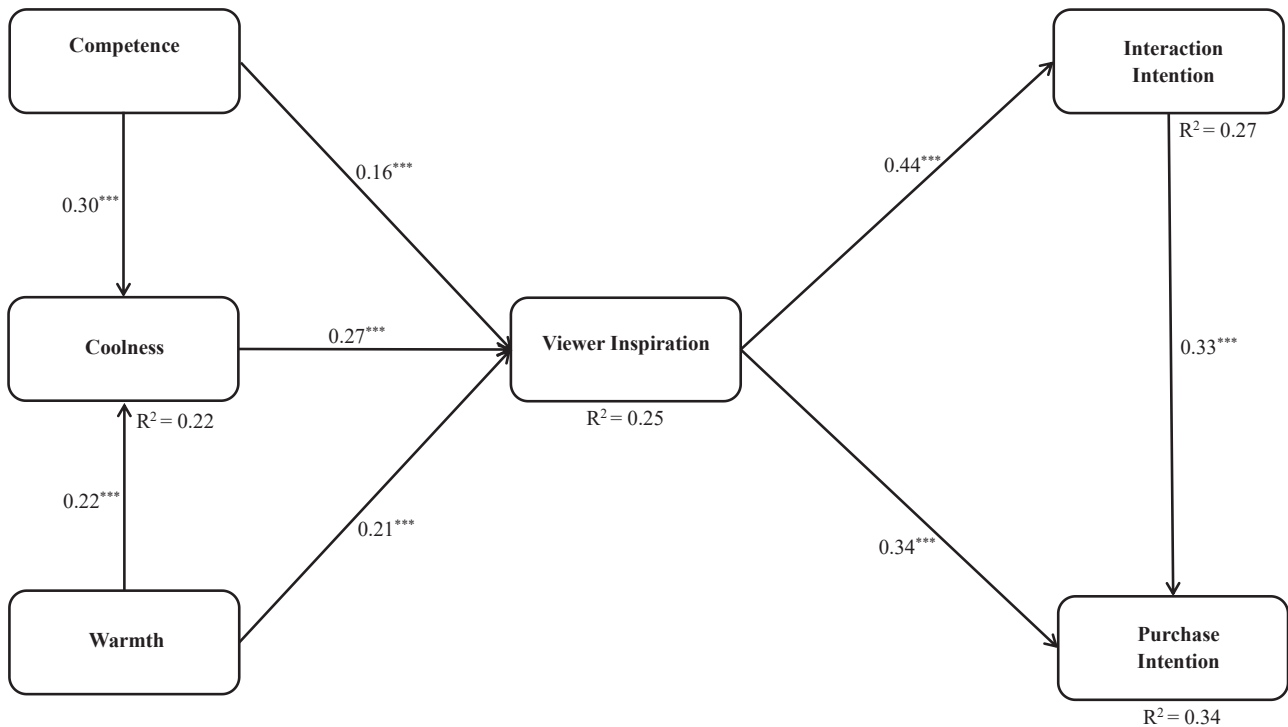


Fig. 2. Results of the research model Notes: Full lines indicate supported hypotheses. *** indicates that the path coefficient is significant at $p < 0.001$.

Table 4

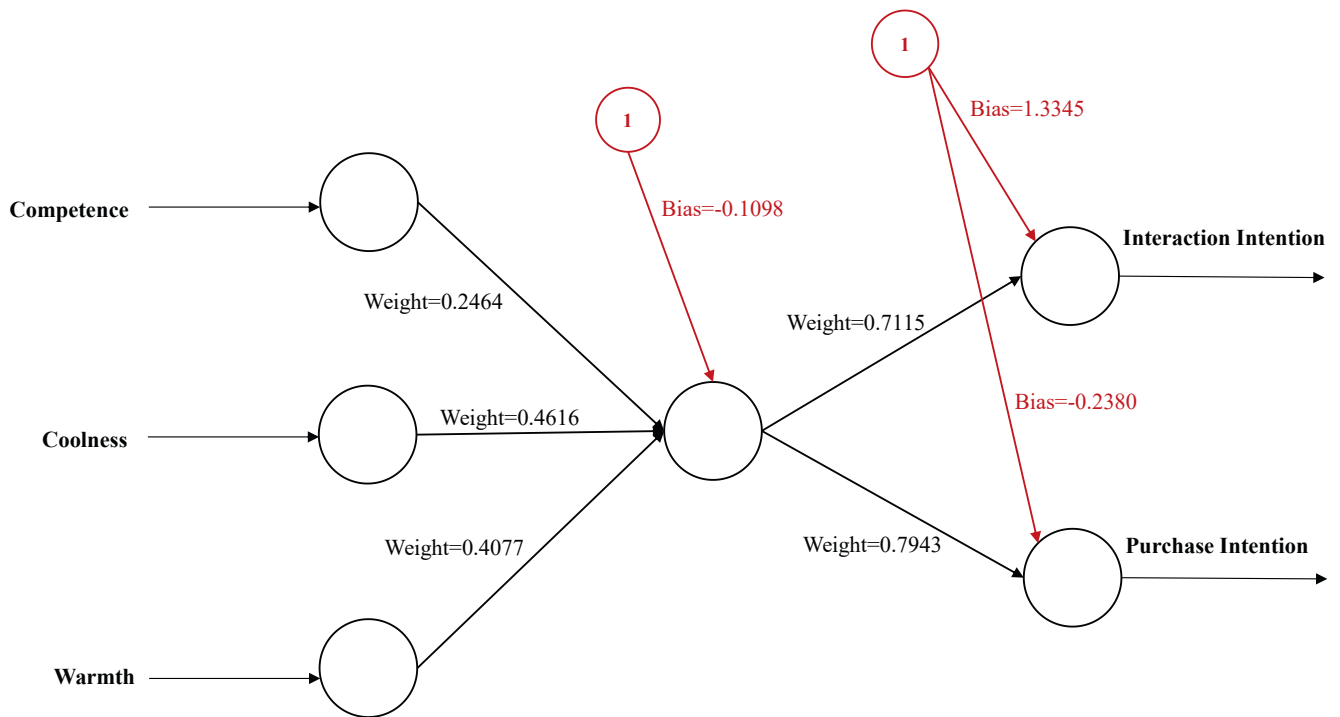
The hypothesized path coefficients.

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H2	Warmth → Coolness	0.22	<0.001	0.14	0.30	0.08	Supported
H3	Competence → Viewer inspiration	0.16	<0.001	0.08	0.24	0.06	Supported
H4	Warmth → Viewer inspiration	0.21	<0.001	0.13	0.30	0.08	Supported
H5	Coolness → Viewer inspiration	0.27	<0.001	0.19	0.35	0.11	Supported
H6	Viewer inspiration → Interaction intention	0.44	<0.001	0.36	0.52	0.22	Supported
H7	Viewer inspiration → Purchase intention	0.34	<0.001	0.26	0.42	0.17	Supported
H8	Interaction intention → Purchase intention	0.33	<0.001	0.25	0.41	0.16	Supported

Table 5

Mediation analysis.

Mediator	Hypothesis	Path	Indirect Effect	p-value	Effect Size	Results
Viewer Inspiration	H9a	Competence → Interaction intention	0.11	<0.001	0.04	Supported
	H9b	Coolness → Interaction intention	0.12	<0.001	0.05	Supported
	H9c	Warmth → Interaction intention	0.12	<0.001	0.05	Supported
	H10a	Competence → Purchase intention	0.12	<0.001	0.04	Supported
	H10b	Coolness → Purchase intention	0.13	<0.001	0.04	Supported
	H10c	Warmth → Purchase intention	0.13	<0.001	0.06	Supported

**Fig. 3.** ANN model.

validated by this model.

As seen by the actual vs forecast plots for interaction intention and purchase intention in Fig. 4 and Fig. 5, respectively, the model's prediction abilities produced good results.

5. Discussion and conclusion

Virtual AI streamers have shown a strong capacity to facilitate interaction and guide shopping in the LSE [98]. However, research in this field is still in its infancy with only studies by Feng et al. [14], Wang et al. [8] and Gao et al., [10]. Therefore, additional research exploration is required to explain viewer response towards such streamers. This study investigated how image dimensions of virtual AI streamers influence its viewers' interaction and purchase intentions via inspiration, and

thus contributes to knowledge in LSE, anthropomorphism, and inspiration and offers suggestions for live-streaming companies and retailers interested in using virtual AI streamers in marketing and promotion.

5.1. Theoretical implications

5.1.1. Influence of virtual AI streamers on viewers

This study depicts the process of how virtual AI streamers affect viewers' behavioral intentions and thus enriches the LSE literature. The results in this study supported all hypothesised relationships. This means, viewers are inspired by the three image dimensions of the AI virtual streamer (competence, warmth, and coolness) to develop interaction and purchase intentions. The influence on viewer behaviors indicates that the impact of virtual AI streamers could be compared to

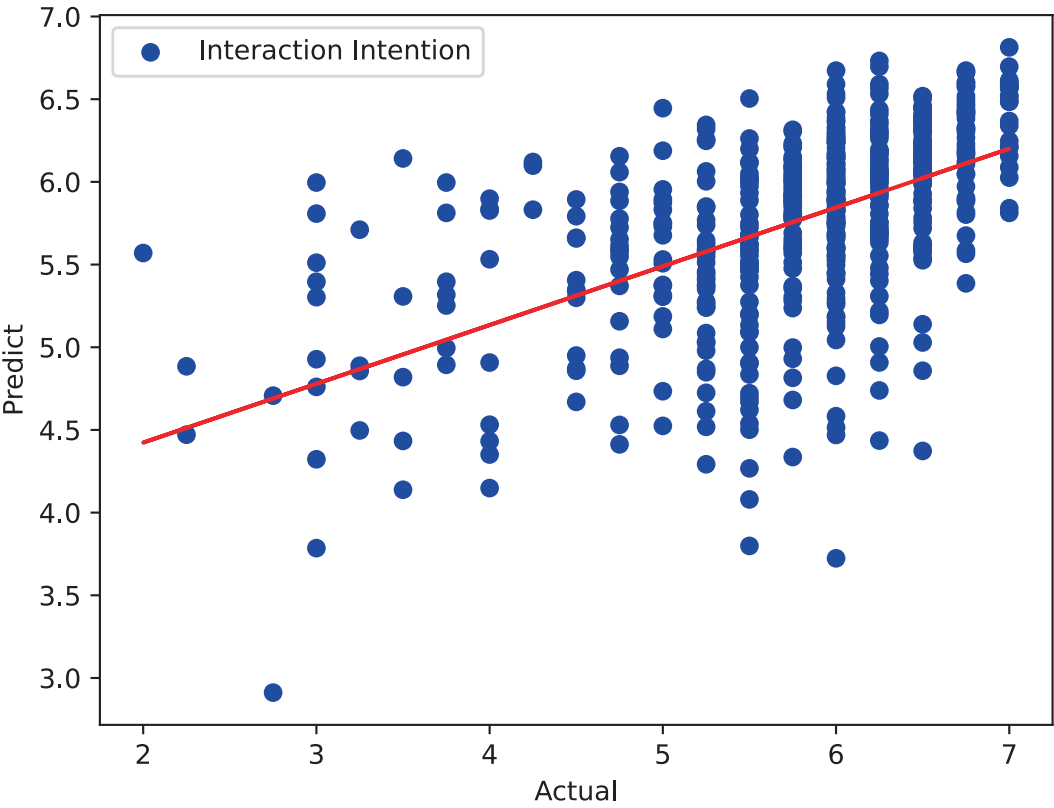


Fig. 4. Actual vs. predicted interaction intention.

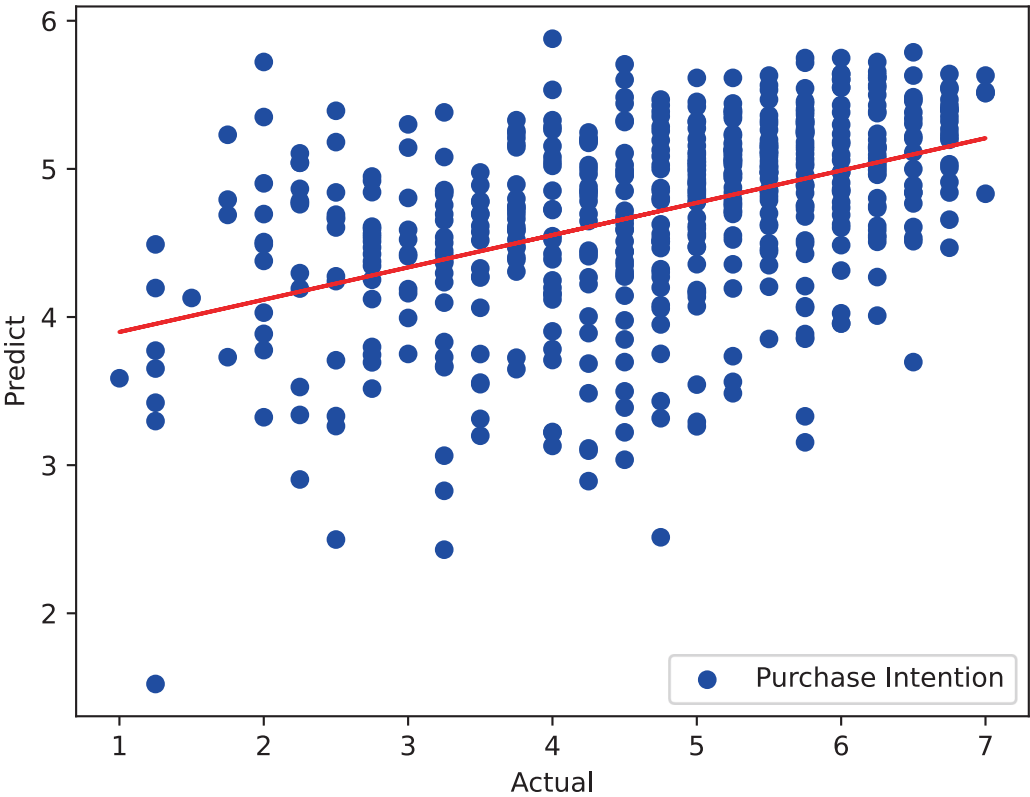


Fig. 5. Actual vs. predicted purchase intention.

human streamers, social media influencers, and even celebrities [99]. However, virtual AI streamers are unique in the influencing mechanism as it is its human-likeness that inspire viewers to act while human influencers rely more on the human attractiveness and influencer-viewer fit to exert influence [100].

5.1.2. Anthropomorphism of virtual AI streamers

This study provides a new perspective on the anthropomorphism in AI. Previous studies have mostly focused on the degree and function of anthropomorphism (e.g., boosting sales, [101]; influencing acceptance and trust, [102]; “uncanny valley” effect, [103]). However, understanding the specific description of the anthropomorphism aspect may be also necessary to the research of AI [104]. Therefore, instead of simply focusing on the previous measurements of the degree it resembles humans, this study explained anthropomorphism as a mix of different human-like elements: competence, warmth, and coolness. Competence facilitates the completion of knowledge-based tasks (e.g., product recommendation and information offering), warmth for emotion-laden tasks (e.g., customer entertainment and companionship) [105], and the uniqueness and innovativeness in coolness might bring viewers new ideas and insights [106].

5.1.3. Inspiration in virtual AI live-streaming

This study also enriches inspiration literature. While previous inspiration studies found its sources included online opinion leaders [107], brand endorsers [108], and AR technology [29], this study extends it to image of virtual AI streamers.

According to the results, although all three image dimensions influence viewer inspiration, competence exhibits the weakest effect, which is opposite to the findings in previous studies on human streamers. Guo et al. [37] found that competence, among five human streamer characteristics including warmth, had the strongest effect on both utilitarian and hedonic value in settings where no particular brand or product type was emphasized. One possible reason for the divergence is that streamers in Guo et al.’s [37] study were product-oriented and acted as shopping advisors, whereas streamers in this study focus on performance and interaction, consistent with popular virtual AI streamers such as Neuro-sama who only briefly introduce products or brands [69]. As a result, competence may be less salient in the context of this study compared to product-driven evaluations [35]. Another plausible reason is different response mechanisms towards AI versus human agents. According to Shank et al. [109], AI is perceived as more powerful than humans but lower in goodness. Therefore, viewers in this study may take competence of virtual AI streamers for granted and respond more strongly to the unexpected socially evocative image of warmth and coolness. This indicates a distinct viewer response pattern when viewing virtual AI streamers compared to human ones.

In terms of the outcomes of inspiration (interaction intention and purchase intention) in this study, the results are consistent with previous research which suggested inspiration affect customer engagement behaviors and purchase intention [110] and this study incorporates virtual AI streamers into the inspiration discussion.

5.2. Practical implications

In the contemporary technology-pervasive market, AI applications are increasingly being implemented in all industries and in the LSE industry, AI technology is now used as virtual AI streamers [68,111]. Brand managers could adopt virtual AI streamers to improve customer engagement and product sales. The findings of this study provide several suggestions for practitioners of LSE to better implement virtual AI streamers to boost product turnover.

To maximize viewer inspiration and subsequently enhance their intention to interact and purchase, developers should develop images of AI virtual streamers to deliver cool, warm, and competent images to viewers, especially coolness which has the strongest effect on

inspiration. “Cool hunting” could be a way to increase the coolness of virtual AI influencers, which is to identify upcoming cool and innovative trends so that developers can design the image, including clothes, makeup, gestures, manner of speaking, based on the cool trends [106]. Besides, they can create a warm, friendly, and considerable image of virtual AI streamers by designing smiling facial expressions and a friendly language style for more interaction. Moreover, developers could enhance the streamers’ ability to portray competence, such as the ability to communicate, handle customer queries, and describe products. For retailers, they should select and modify the image of virtual AI streamers for their brand building according to the results, so that the viewers of the retailers’ live-streaming are likely to be inspired by such streamers to purchase the retailers.

5.3. Limitations and future research

First, this research scenario was set up to simply provide participants with text descriptions, pictures, and a video of the AI virtual streamer without actually showing them the real live stream. Future research could use other methods such as experiments that allow participants to watch a complete AI live stream and record their behaviors during the viewing period, asking them about their mental activity. Second, although participants without previous live-streaming shopping experience were screened out, their prior experience of watching virtual AI streamers was not assessed. Participants lacking such experience may have difficulty in fully imagining the virtual AI streamer scenario, which may reduce response accuracy and limit the generalizability of findings to real-world viewers familiar with virtual AI streamers [112]. This may explain the relatively high and narrowly ranged mean scores on image dimensions (ranging from 5.29 to 5.99) as participants that lack concrete references may have defaulted to generic or socially desirable judgments [113]. It may also account for the relatively high mean scores of inspiration (5.51) as inspiration is likely to be activated by the novel stimulus as already mentioned. Future studies could include prior experience of watching virtual AI streamers as a control variable to improve response accuracy and generalizability of findings. Third, this survey did not specify product types in the commerce show in order to avoid any potential external effect; respondents’ intentions may vary depending on the product type, however. Future research could investigate whether product categories, brand information, or price-related factors influence viewers’ responses to virtual AI streamers.

In addition, this study only examined the mediating role of “inspiration by” in the customer inspiration concept, which may limit the comprehensiveness of this concept. Thus, further research could include the twin components of customer inspiration (i.e., “inspired by” and “inspired to”) proposed by Böttger et al. [39] and examine the relationships between them. Further research to investigate other antecedents and consequences of customer inspiration is also significant. Moreover, future studies may include variables that moderate the effect of customer inspiration on behavioral responses such as cultural orientation (e.g., individualism/collectivism) to extend the model [114].

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CRediT authorship contribution statement

Ting (Tina) Li: Writing – review & editing, Conceptualization, Writing – original draft. **Zhongyuan Zhou:** Data curation, Methodology, Formal analysis. **Xianfeng Zhang:** Writing – review & editing, Funding acquisition, Supervision. **Yang Zhou:** Project administration. **Si Wen:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.101000>.

Data availability

Data will be made available on request.

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